ELECTRICAL MEASUREMENTS

Dr. Eng. / Mohamed I. Abdo
Lecturer - Industrial Electronics and Control Eng. Dept.
Faculty of Electronic Engineering - Menoufia University

ELECTRICAL MEASUREMENTS

Scope of presentation

- **Introduction**
- **Sources of systematic error**
  1. **System disturbance due to measurement**
  2. **Errors due to environmental inputs**
  3. **Wear in instrument components**
  4. **Connecting leads**
- **Reduction of systematic errors**
- **Random errors**
- **Statistical analysis of measurements subject to random errors**
- **Aggregation of measurement system errors**
- **Thanks & Questions**

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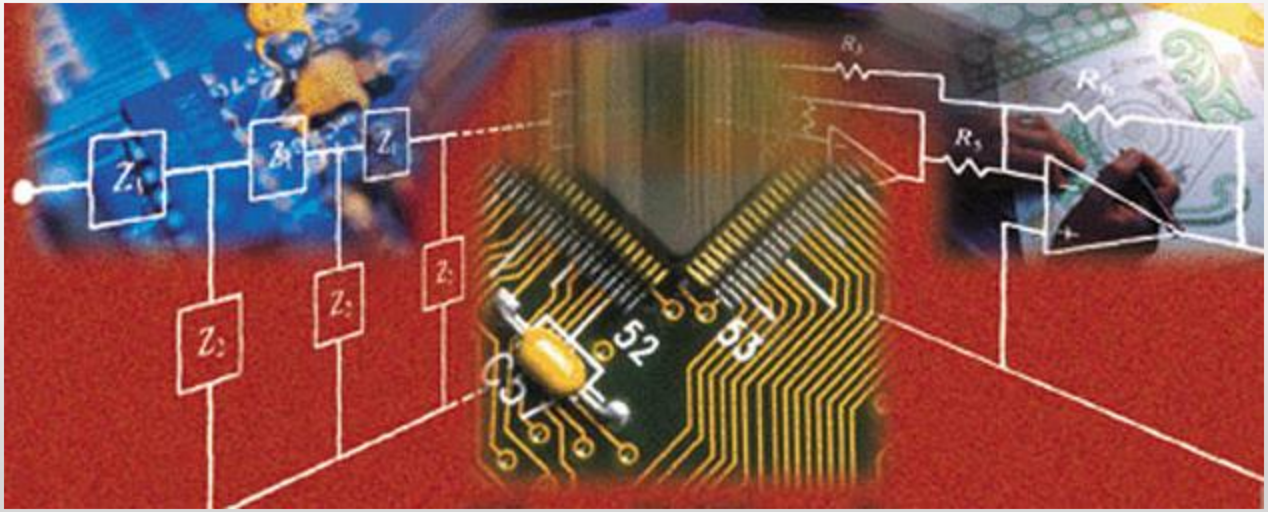


ERRORS DURING THE MEASURMENT PROCESS

- **Errors in measurement systems** can be divided into

1. those that arise during the measurement process.
2. and those that arise due to later corruption of the measurement signal by induced noise during transfer of the signal from the point of measurement to some other point.

This chapter considers only the first of these,

ERRORS DURING THE MEASURMENT PROCESS

➢ Error Sources

✓ Errors arising during the measurement process

- can be divided into two groups, known as systematic errors and random errors.

1. Systematic errors describe errors in the output readings of a measurement system that are consistently on one side of the correct reading, i.e. either all the errors are positive or they are all negative.

ERRORS DURING THE MEASURMENT PROCESS

➢ Error Sources

✓ Errors arising during the measurement process

- can be divided into two groups, known as systematic errors and random errors.

2. Random errors are perturbations of the measurement either side of the true value caused by random and unpredictable effects, such that positive errors and negative errors occur in approximately equal numbers for a series of measurements made of the same quantity, when measurements are taken by human observation of an analogue meter and Electrical noise.

Sources of systematic error

➢Sources of systematic error

1. System disturbance due to measurement.
2. Errors due to environmental inputs.
3. Wear in instrument components.
4. Connecting leads.

These various sources of systematic error, and ways in which the magnitude of the errors can be reduced, are discussed below.

Sources of systematic error

1. System disturbance due to measurement.

- The magnitude of the disturbance varies from one measurement system to the next and is affected particularly by the type of instrument used for measurement.
- Ways of minimizing disturbance of measured systems is an important consideration in instrument design. However, an accurate understanding of the mechanisms of system disturbance is a prerequisite for this.

Sources of systematic error

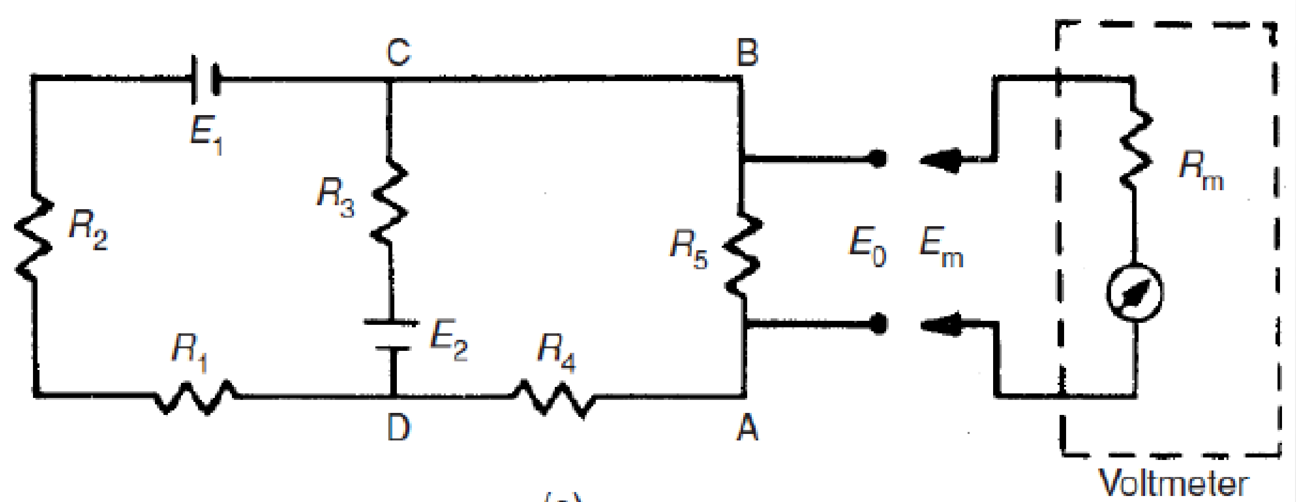
1. System disturbance due to measurement.
   - ✓ Measurements in electric circuits

- In analysing system disturbance during measurements in electric circuits, Thevenin's theorem is often of great assistance.

- For instance, consider the circuit shown in Figure 3.1(a) in which the voltage across resistor R5 is to be measured by a voltmeter with resistance Rm.

Sources of systematic error

1. System disturbance due to measurement.
   - ✓ Measurements in electric circuits



(a)

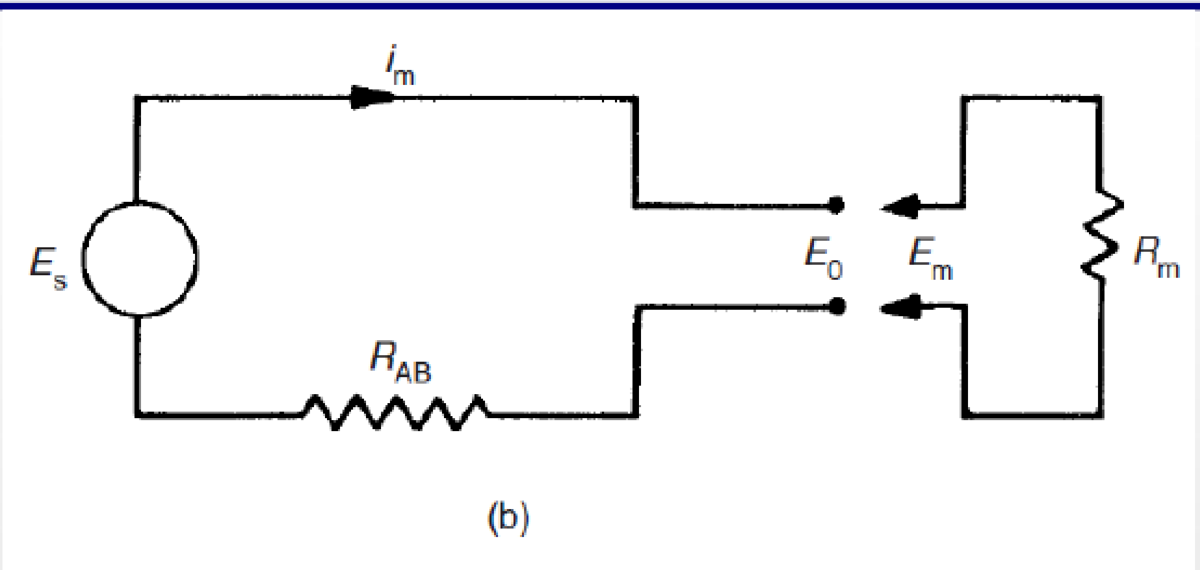
Sources of systematic error

1. **System disturbance due to measurement.**
   - ✓ Measurements in electric circuits

- Here, Rm acts as a shunt resistance across R5, <u>decreasing the resistance between points AB and so disturbing the circuit.</u>
- Therefore, the voltage Em measured by the meter is not the value of the voltage E0 that existed prior to measurement.
- The extent of the disturbance can be assessed by calculating the open circuit voltage E0 and comparing it with Em.

Sources of systematic error

1. System disturbance due to measurement.
   - ✓ Measurements in electric circuits



(b) Equivalent circuit by Thevenin's theorem

Sources of systematic error

1. System disturbance due to measurement.

✓ Measurements in electric circuits

$$\frac{1}{R_{CD}} = \frac{1}{R_1 + R_2} + \frac{1}{R_3} \quad \text{or} \quad R_{CD} = \frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3}$$

$$\frac{1}{R_{AB}} = \frac{1}{R_{CD} + R_4} + \frac{1}{R_5} \quad \text{or} \quad R_{AB} = \frac{(R_4 + R_{CD})R_5}{R_4 + R_{CD} + R_5}$$

$$R_{AB} = \frac{\left[\frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3} + R_4\right] R_5}{\frac{(R_1 + R_2)R_3}{R_1 + R_2 + R_3} + R_4 + R_5}$$

$$I = \frac{E_0}{R_{AB} + R_m}, \quad E_m = \frac{R_m E_0}{R_{AB} + R_m}.$$

$$\frac{E_m}{E_0} = \frac{R_m}{R_{AB} + R_m}$$

Sources of systematic error

1. **System disturbance due to measurement.**
   - ✓ **Measurements in electric circuits**

$$\frac{E_m}{E_0} = \frac{R_m}{R_{AB} + R_m}$$

- **It is thus obvious that as Rm gets larger, the ratio Em/E0 gets closer to unity,**
- **showing that the design strategy should be to make Rm as high as possible to minimize disturbance of the measured system.**
- **(Note that we did not calculate the value of E0, since this is not required in quantifying the effect of Rm.)**

End of Lecture

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Good Luck

See You in Next Lecture , , ,

With My Best Wishes

Dr. Eng. Mohamed Abdo